Preparation, calibration and exact-record fibres of a classical apparatus

For the finite autonomous apparatus, exact final records can hide a whole receiver shell when incoming apparatus phase is unknown. Two initial energies and one or two probe-position calibrations still permit actual common-record canonical alternatives in selected fixed designs. Three calibrated positions and apparatus energy instead give global recovery on a sufficiently small fixed box. Finite calibration tolerance restores ambiguity with matched risk-product order $\min\{1, (\varepsilon/\lambda)^2\}$. The direction of uncertainty matters: a correlated line can hide only an internal coordinate at leading order, and its first canonical tangent cancels exactly.

These results consolidate C090–C093, C096–C101 and C106–C123 by their common inverse construction. The apparatus and its smooth-flow estimates are specified once in the readout treatment. Historical written coordinator/model checks are internal checks, not external peer review; literature novelty is unassessed. Distinct rank, topology and energy hypotheses below must not be inferred from one another.

Common apparatus chart and the meaning of risk

Use the receiver $z = (x, P, y, Q)$, energy $H_s = z^T G z/2$, clock and four probes of the readout treatment. Put $k_x = a - g^2/d > 0$. All norms use fixed component units L_*, P_* for the canonical receiver coordinates. Let η be all ten initial apparatus coordinates, $G_\lambda(z, \eta)$ their ten final coordinates at known T, and S the free apparatus shear

$$(q_j, \pi_j, s, p_s) \mapsto (q_j + T\pi_j/M_j, \pi_j, s + Tp_s/M_c, p_s).$$

On convex compact receiver/apparatus enlargements with strict cutoff and pulse margins, smooth finite-time flow gives

$$F_\lambda := S^{-1}G_\lambda = \eta + \lambda R_\lambda(z, \eta), \qquad \|D_\eta F_\lambda - I\| \le C\lambda \le \tfrac{1}{2}, \quad \|D_z F_\lambda\| \le L\lambda.$$

Derivatives of every finite order used below are bounded on smaller compact domains, including a small signed coupling interval. At a prescribed record $a = S^{-1}Y$, the map $\eta \mapsto a - \lambda R_\lambda(z, \eta)$ is a contraction on an enlarged apparatus ball for a near its centre. It defines a unique smooth inverse apparatus chart $\eta_\lambda(z, a)$, with $\eta_0(z, a) = a$. If $Y = G_\lambda(z_*, \eta_*)$, then

$$G_\lambda(w, \eta_\lambda(w)) = Y, \qquad \|\eta_\lambda(w) - \eta_*\| \le 2L\lambda\|w - z_*\|.$$

Indeed $\eta \mapsto \eta - F_\lambda(w, \eta) + F_\lambda(z_*, \eta_*)$ contracts the radius-$b/2$ ball about η_* into itself whenever $\|w - z_*\| \le b/(4L\lambda)$. Its centre moves by at most $b/4$, its derivative norm is at most $1/2$, and its fixed point has the displayed bound. This includes exact nonlinear clock reaction. Counterfactual chart values may leave a smaller physical box while remaining in the enlarged smooth domain; segment proofs use this extension explicitly.

For a deterministic estimator let ϵ_x, ϵ_P be separate global worst-case errors. A common-record pair with coordinate separations $\Delta x, \Delta P$ forces $\epsilon_x \ge |\Delta x|/2$ and $\epsilon_P \ge |\Delta P|/2$. These need not be errors on the same motion. A smooth curve may have positive coordinate-risk product but zero projected planar area. All products here have action units ML^2/T, without a 2π factor, and are reconstruction risks, not accuracy–disturbance inequalities. Correlated choices used in a lower bound select points from the admitted Cartesian support; no probability law or positive probability for a fibre is assumed.

Incomplete records and preparation width

First retain only four final probe momenta. Unknown incoming probe momenta u have derivative $D_u G = I + O(\lambda)$, while $D_z G = O(\lambda)$. The same contraction hides every shell point within

$b/(4L\lambda)$ of z_*. At $z_* = (0, 0, \sqrt{2E/d}, 0)$ the exact shell chart is

$$w(x, P) = \left(x, P, \frac{g}{d}x + \sqrt{\frac{2E - k_x x^2 - P^2/\mu}{d}}, 0 \right).$$

Choose $r_0, C_0 > 0$ so its square $|x| \le L_* r_0$, $|P| \le P_* r_0$ has positive radicand and $\|w - z_*\| \le C_0 \max(|x|/L_*, |P|/P_*)$. Then one record hides the whole square with

$$r = \min\{r_0, b/(4LC_0\lambda)\}, \qquad \epsilon_x \epsilon_P \ge L_* P_* r^2, \quad \text{area}_{x,P} \ge 4L_* P_* r^2.$$

At fixed coupling the small-width rate is b^2, matching the calibrated upper bound. At fixed width the displayed lower bound saturates as coupling weakens. No exact total apparatus energy has yet been imposed.

If incoming momenta are revealed at zero but incoming positions q remain unknown, the response has two coupling factors. With $h(t) = e_x \Phi_t e_P$, $h(0) = 0$, $h'(0) = 1/\mu$, and $f_j(t) = f_j(s_0 + v_0 t)$, its exact leading matrix is

$$D_q G_\lambda = \lambda^2 B(z) + O(\lambda^3),$$
$$B_{jk} = K^2 \int_0^T f_j(t) \int_0^t h(t-u) f_k(u) du dt + \frac{K^2}{M_c} \int_0^T f_j'(t) x^0(t) \int_0^t (t-u) f_k'(u) x^0(u) du dt.$$

It follows by differentiating receiver and clock equations in q_k: $\partial_{q_k} x = -\lambda K \int_0^t h(t-u) f_k(u) du + O(\lambda^2)$ and $\partial_{q_k} s = -\lambda K M_c^{-1} \int_0^t (t-u) f_k'(u) x^0(u) du + O(\lambda^2)$. In particular the clock term cannot be discarded. Ordered disjoint pulses make B lower triangular. Select independent signal times avoiding the isolated zeros of the nonzero analytic history at z_*. For $f_j(t) = \phi((t - t_j)/\delta)$, fixed smooth nonzero $\phi \ge 0$,

$$B_{jj} = -\frac{K^2 x^0(t_j)^2 \delta}{M_c v_0^2} \int \phi^2 + O(\delta^2).$$

The receiver term is $O(\delta^3)$. For the clock term substitute scaled pulse variables: the inner integral of $(r - s)\phi'(s)$ is $\int_{-\infty}^r \phi$, and integration by parts gives $-\int \phi^2$. Choose sufficiently small positive δ once, making B_* invertible. Preconditioning by $(\lambda^2 B_*)^{-1}$ gives a contraction when $\|w - z_*\| \le \lambda b/(4C)$, with $C \ge \|B_*^{-1}\| L$. The same shell square now has

$$r = \min\{r_0, \lambda b/(4CC_0)\}, \qquad \epsilon_x \epsilon_P \ge L_* P_* r^2, \quad \text{area}_{x,P} \ge 4L_* P_* r^2.$$

This is a design-existence result, not a consequence of ordinary signal rank for every pulse design. Its lower bound closes as λ^2 at fixed b; that does not prove the full risk tends to zero. Complete final pointer positions with known clock data instead give global recovery by the readout inverse.

Full records, whole-shell saturation and exact energies

With the entire incoming apparatus box unknown, the common ten-record chart at the start applies without any signal-rank assumption. On $H_s = E$ let $D = \max \|w - z_*\|$. If

$$0 < \lambda \le \min\{\lambda_0, b/(4LD)\},$$

one complete record hides the entire shell, with a $b/2$ box margin. Its canonical projection is exactly $k_x x^2 + P^2/\mu \le 2E$: completing the energy square proves necessity, and the shell chart with the remaining energy in y proves sufficiency. The constant estimator $(0, 0)$ attains both coordinate lower bounds, hence

$$R_x = \sqrt{2E/k_x}, \quad R_P = \sqrt{2\mu E}, \quad \inf \epsilon_x \epsilon_P = 2E\sqrt{\mu/k_x}.$$

The projected area is $\pi R_x R_P$. The stiffness k_x is an energy projection coefficient, not asserted to be a normal-mode stiffness. At fixed λ the local patch closes with b; the saturation condition cannot then be kept as $b \to 0$.

Supplying apparatus energy $H_0 = M_c v_0^2/2$ removes many compensators, but not all ambiguity. Prescribe terminal zero probes, clock position $s_0 + v_0 T$, and

$$z(T) = r\Phi_T z_*, \qquad p_s(T) = \sqrt{2M_c[H_0 + E(1-r^2)]}.$$

Backward exact flow has initial receiver $Z_\lambda(r)$ with $Z_0(r) = rz_*$. The derivative of $H_s(Z_0(r)) - E$ at $r = 1$ is $2E > 0$, so an implicit r_λ makes initial receiver energy exactly E. Endpoint interactions vanish; total-energy conservation makes apparatus energy exactly H_0. Terminal zero probes give initial probe phase $O(\lambda)$ and receiver/clock reaction $O(\lambda^2)$, so $r_\lambda = 1 + O(\lambda^2)$ and $Z_\lambda = z_* + O(\lambda^2)$. At fixed positive box width both preparations fit within half its margin.

Inside the linear cutoff regions $(z, q, \pi) \mapsto (-z, -q, -\pi)$ leaves the clock and equations invariant. Terminal zero probes therefore give the same complete apparatus record for Z_λ and $-Z_\lambda$, with both energies equal. Global oddness of cutoffs is unnecessary. Choose $z_* = (\sqrt{E/k_x}, \sqrt{\mu E}, (g/d)\sqrt{E/k_x}, 0)$. Then

$$\epsilon_x \epsilon_P \geq |(Z_\lambda)_x (Z_\lambda)_P| \longrightarrow E\sqrt{\mu/k_x} > 0.$$

For small coupling the bound is at least one quarter of this limit. This two-point construction asserts neither exact minimax saturation nor positive area. Its positive scale closes with receiver energy.

Clock fibres and local versus global recovery

Keep incoming momenta known at zero and only eight final pointer records. Their scaled map is $(-A_c z, q) + O_{C^1}(\lambda)$. The implicit-function theorem solves equal records as

$$z_0(c) = A_c^{-1} A_{c_0} z_*, \quad q_0(c) = 0, \qquad H_s(z_\lambda(c)) = E.$$

Interior pulse support gives $\partial_s A = -AM/v$ by integration by parts. Thus $\partial_s z_0 = Mz_*/v_0$ and the offset derivative of energy is zero. To obtain a transverse speed direction choose early pulses $f_j(s) = \phi((s - s_0 - v_0\epsilon a_j)/(v_0\epsilon\eta))$, with distinct positive a_j and fixed sufficiently small $\eta > 0$. The moment matrix $W_{jn} = \int \phi((u - a_j)/\eta) u^n/n!\, du$, $n = 0, 1, 2, 3$, is invertible by its Vandermonde limit. With $r = v_0/v$,

$$\det A_{s_0,v} = D(r\epsilon)^{10}(1 + O(\epsilon)), \quad D \neq 0, \qquad \operatorname{tr}(-A^{-1}\partial_v A) = 10/v_0 + O(\epsilon) > 0.$$

The exponent is four time integrations plus derivative orders $0 + 1 + 2 + 3$; the analytic expansion holds also with one speed derivative. Fix sufficiently small positive ϵ once. Positive trace implies the symmetric energy form of $B = -A^{-1}\partial_v A$ has a positive direction: its G-normalized trace equals $\operatorname{tr} B$. Select a shell point in its open positive set avoiding $P = 0$ and $-ax + gy = 0$. Then $z_*^T GBz_* > 0$ and both canonical entries of Mz_* are nonzero. Solve energy for $v = v_\lambda(s)$:

$$v_\lambda'(s_0) = O(\lambda), \qquad z_\lambda'(s_0) = Mz_*/v_0 + O(\lambda).$$

On a fixed offset interval $|s - s_0| \leq \sigma$ the two canonical derivatives retain signs and magnitudes at least $k_x', k_P' > 0$. Speed/position compensations are $O(\lambda\sigma)$ and fit fixed positive boxes. Its common-record endpoints imply $\epsilon_x \epsilon_P \geq k_x' k_P' \sigma^2 > 0$. This curve has no implied positive projected area. Revealing offset alone removes this family locally through the transverse energy derivative, as proved in the local inverse.

On the whole shell, two speed branches remain possible. Factor the early pulse matrix before inversion, with O having rows $e_x M^n$:

$$-A^{-1}\partial_v A = \frac{1}{v_0} O^{-1}\,\mathrm{diag}(1,2,3,4)O + O(\epsilon).$$

The factored analytic tail and its derivative are $O(\epsilon)$ after removing the fixed moment matrix and powers of $r\epsilon$; a vanishing determinant estimate alone would not justify this inverse limit. Its leading map sends z to $(x, 2P, 3y - 2ax/g, 4Q - 2a\nu P/(g\mu))/v_0$. At $Q = 0$ its energy form is

$$v_0 z^T G B^{(0)} z = g(x-y)(9x-5y) + 2P^2/\mu.$$

Choose $L, p > 0$ with $2p^2/\mu < 4gL^2/5$. The normalized shell path through $(L, p, rL, 0)$, $0 \le r \le 7/5$, has opposite endpoint energy-response signs and positive canonical coordinates. Fix the design and then a sufficiently small speed separation $d_v > 0$. At zero coupling equal records require $z' = A^{-1}_{v_0+d_v} A_{v_0} z = z + d_v B z + O(d_v^2)$; the energy mismatch changes sign along the path while both canonical separations exceed positive $c_x d_v, c_P d_v$. At positive coupling the invertible (z', q') record derivative continues this compact path uniformly with $O(\lambda)$ changes. Endpoint signs and margins persist; intermediate value gives an actual exact equal-energy, equal-record pair. For small coupling

$$\epsilon_x \epsilon_P \ge c_x c_P d_v^2/16 > 0.$$

This is an actual pair, not an inference from a zero tangent. Reading final clock momentum instead restores global recovery; reading its final position also replaces initial offset. Those inverse theorems and their precision constants are in the readout treatment.

Calibration constraints reduce to receiver rank

Fix a nonzero initial displacement $q_1 = c$, small enough to lie within one quarter of the original apparatus box. Let η_c have this displacement, all other probe coordinates zero, and nominal clock phase. The inverse apparatus chart centred there compensates any receiver in a bounded domain for sufficiently small coupling relative to the box width.

Before using pulse rank, there is a design-independent full-state obstruction under $H_s = E$, $H_{\mathrm{app}} = H_0$, $q_1 = c$. Solve

$$p_s = \sqrt{2M_c(H_0 - \sum_j \pi_j^2/(2M_j))}, \qquad y = \frac{g}{d}x + \sqrt{(2E - k_x x^2 - P^2/\mu - Q^2/\nu)/d}.$$

The free chart coordinates are $x, P, Q, q_2, q_3, q_4, \pi_1, \ldots, \pi_4, s$, eleven in total. A fixed radius-r Euclidean ball fits with strict radicands and half-box margins. Its boundary S^{10} maps continuously to the ten final records. Borsuk–Ulam gives antipodal chart parameters with equal records. Equal-record ambient estimates imply $\|\Delta\eta\|_2 \le 2L\lambda\|\Delta z\|_2$. Since chart coordinates are an orthogonal projection of fixed-unit initial coordinates,

$$R_z \ge \frac{r}{\sqrt{1 + 4L^2\lambda^2}} > 0.$$

This is a dimensionless full receiver-state risk. Separation may be in y, Q; it does not prove a canonical product. The chart antipodes do not reverse c, so the broken sign involution is not being reused.

For canonical constructions set $Y_\lambda = G_\lambda(\bar{z}, \eta_c)$ and write $\eta_\lambda(w)$ for its compensator. Define

$$A_j(w) = \int_0^T t f_j(s_0 + v_0 t) x_w(t) dt, \qquad B_1(w) = \int_0^T f_1(s_0 + v_0 t)\dot{x}_w(t) dt.$$

Dividing calibration and energy residuals by coupling gives smooth limits

$$\frac{q_j(\eta_\lambda(w)) - q_j(\eta_c)}{\lambda} \longrightarrow -\frac{K}{M_j} A_j(w - \bar{z}), \qquad \frac{H_{\mathrm{app}}(\eta_\lambda(w)) - H_0}{\lambda} \longrightarrow -KcB_1(w - \bar{z}).$$

The exact drift identity is $q_j(T) - T\pi_j(T)/M_j = q_j(0) + (\lambda K/M_j) \int t f_j(s) x dt$. Matching final clock momentum gives the initial clock-impulse difference; its energy variation is v_0 times that impulse. Integration by parts turns $\int f_1' x dt$ into $-\int f_1 \dot{x} dt / v_0$, proving the energy row and its sign. Initial zero probe momenta have no first-order kinetic-energy variation. For every smooth numerator N vanishing at zero coupling, $N/\lambda = \int_0^1 \partial_\lambda N(\theta\lambda, \cdot) d\theta$, which proves joint smooth divisibility, including receiver derivatives.

With only $q_1 = c$, a narrow fixed pulse near a small $\tau > 0$ makes the two rows A_1, B_1 approximate $x(\tau), \dot{x}(\tau)$ after nonzero normalization. Their kernel contains a vector close to $\Phi_{-\tau}(0, 0, Y, 0)$, with

$$v_x = gY\tau^2/(2\mu) + O(\tau^4), \qquad v_P = -gY\tau + O(\tau^3),$$

both nonzero. In their two-dimensional kernel W, choose $\bar{z} \neq 0$ G-orthogonal to v, scaled to $H_s = E$. Then $dH_s(\bar{z}), A_1, B_1$ have rank three and common kernel $\mathbb{R}v$. Adding a parameter functional $\ell(v) = 1$ gives an invertible four-equation implicit system. It continues an actual common-record shell curve $w_\lambda(t)$, $|t| \leq \delta$, with $w_\lambda(0) = \bar{z}$ and $\partial_t w_0(0) = v$. One fixed compact parameter/coupling rectangle retains half the two canonical derivative magnitudes and half-box apparatus margins. Its endpoints therefore give

$$\epsilon_x \epsilon_P \geq \delta^2 |v_x v_P|/4 > 0.$$

This pulse design can retain the ordinary four-signal matrix: analytic observability completes the first evaluation row with three disjoint times. The result is existence within the design class, not every pulse design.

Adding $q_2 = 0$ gives the third row A_2. Pulses near $\tau, 2\tau$ have a rank-three kernel obtained from state $(0, 0, Y, Q)$ at τ, with $Q > 0$ and $Y = -Q\tau/(3\nu) + O(\tau^3)$ to make $x(2\tau) = 0$. Backward expansion gives

$$v_x = -gQ\tau^3/(3\mu\nu) + O(\tau^5), \qquad v_P = 5gQ\tau^2/(6\nu) + O(\tau^4).$$

Fix positive widths preserving both nonzero entries and rank. Scale v to $\bar{z}$ on the shell. At zero coupling the three linear constraints and energy have two roots $\pm\bar{z}$, with invertible derivatives because $dH_s(\pm\bar{z})[\pm\bar{z}] = 2E$. The exact compensated system continues both roots as $w_+ = \bar{z}$, $w_- = -\bar{z} + O(\lambda)$. They have the same full record, energies and calibrations, with margins, and

$$\epsilon_x \epsilon_P \geq \tfrac{1}{4} |(w_+ - w_-)_x (w_+ - w_-)_P| \longrightarrow |\bar{z}_x \bar{z}_P| > 0.$$

Each root is locally stably recoverable at each fixed positive coupling: eliminate the invertible apparatus-to-record derivative in the augmented fourteen-coordinate map; its Schur derivative is the four-equation residual derivative, with the three divided rows multiplied by λ. It remains invertible. A sufficiently small convex neighbourhood gives a local lower Lipschitz estimate. These neighbourhoods/constants need not be uniform as $\lambda \to 0$. Two local inverses do not give global recovery.

Four exact supplied constraints give a global inverse

Take a bounded convex receiver energy ball $Z = \{H_s \le E_{\max}\}$ and $C(\eta) = (q_1, q_2, q_3, H_{\mathrm{app}})$, with supplied $C_* = (c, 0, 0, H_0)$. Choose the third pulse nonzero on the preceding one-dimensional kernel, and a fourth completing the ordinary signal matrix. Analytic observability on any remaining interval guarantees both choices; choose positive widths preserving them. In fixed output units the matrix

$$L = \begin{pmatrix} -KA_1/M_1 \\ -KA_2/M_2 \\ -KA_3/M_3 \\ -KcB_1 \end{pmatrix}$$

is invertible. Nonzero c is essential. The physical apparatus box is now centred at η_c with sufficiently small fixed half-width $\rho > 0$, chosen after design. This may be smaller than the preceding two-branch box; no recovery theorem on that original larger box is claimed.

For all nearby records use the enlarged inverse apparatus chart and define

$$D_\lambda(w, a) = [C(\eta_\lambda(w, a)) - C(a)]/\lambda.$$

Subtracting $C(a)$ is necessary for joint smoothness when the record varies. At $a = \eta_c$ its zero-coupling receiver derivative is exactly L for every $w \in Z$. Uniform continuity permits a chart radius and then coupling ceiling such that

$$\|L^{-1}(D_w D_\lambda - L)\| \le \tfrac{1}{2}.$$

Choose ρ inside that chart (for example at most one eighth the enlarged radius), so all physical records and their segments stay in its domain. Receiver segment integration gives

$$\|w - w'\| \le 2\|L^{-1}\|\|D_\lambda(w, a) - D_\lambda(w', a)\|.$$

For equal records and exact $C = C_*$ the right side is zero. This proves uniform global recovery over the whole bounded domain, without supplying receiver energy or a local branch label. For different records, bounds $M_C \ge \|DC\|$, $M_D \ge \|D_a D_\lambda\|$ give

$$\|w - w'\| \le 2\|L^{-1}\|(M_C/\lambda + M_D)\|S^{-1}\|\|Y - Y'\| \le K_z\|Y - Y'\|/\lambda.$$

The apparatus chart also bounds its own error by $K_\eta(\|Y - Y'\| + \lambda\|w - w'\|)$. Minimum-residual fitting on the compact admitted class with record error δ yields $\epsilon_x \epsilon_P \le 4L_* P_* K_z^2(\delta/\lambda)^2$. This is an upper bound with supplied exact calibration and exact model.

Tolerances, exact ambiguity and matched crossover

Allow a reported C_{obs} with $\|C(\eta) - C_{\mathrm{obs}}\| \le \varepsilon$, independently of final-record error δ. The same inverse comparison, using actual C_1, C_2, gives

$$\|w - w'\| \le \frac{K}{\lambda}(\|C_1 - C_2\| + \|Y - Y'\|).$$

Choose any compact feasible preparation consistent with both data bands; it exists because the true preparation is feasible (sequential coordinate minimization makes a deterministic choice). Thus

$$\|\widehat{w} - w\| \le 2K(\varepsilon + \delta)/\lambda, \qquad \epsilon_x \epsilon_P \le 4L_* P_* K^2((\varepsilon + \delta)/\lambda)^2.$$

This covers separate component tolerances via their maximum. The matching ball below allows positive tolerance in all four components; some exactly known components require the directional test in the next section.

At the reference record $Y_\lambda = G_\lambda(0, \eta_c)$, differentiate the inverse chart: $D_w\eta_\lambda = -(D_\eta F_\lambda)^{-1} D_w F_\lambda$. It follows that $\|\eta_\lambda(w) - \eta_c\| \le B\lambda\|w\|$, $B = 2L$. Set $R_Z = \max_Z \|w\|$ and $M_C > 0$. Reduce the fixed coupling ceiling until $B\lambda R_Z \le \rho/2$, also retaining actual q_1 away from zero. Then the same complete record and reported C_* hide the entire dilated receiver ball tZ, where

$$t = \min\{1, \varepsilon/(M_C B\lambda R_Z)\}.$$

Indeed its calibration difference is at most $M_C B\lambda\|w\|$. With $X_E = \sqrt{2E_{\max}/k_x}$ and $P_E = \sqrt{2\mu E_{\max}}$, common-record coordinate endpoints force $\epsilon_x \ge tX_E$, $\epsilon_P \ge tP_E$. For exact final records, combine the feasible-estimator upper bound with the constant-zero estimator in each coordinate. The optimal product has matched order

$$\min\{1, (\varepsilon/\lambda)^2\}$$

with fixed positive action-unit constants. At $\varepsilon = 0$ exact recovery holds. If $\varepsilon \ge M_C B\lambda R_Z$, the whole ball is hidden and $R_x = X_E$, $R_P = P_E$, $\inf \epsilon_x\epsilon_P = X_E P_E$ exactly; its projected ellipse has area $\pi X_E P_E$. This is a sufficient saturation threshold, not a sharp crossover constant. At fixed coupling improved calibration/record precision closes the product; at fixed positive tolerance weak coupling saturates it. Receiver energy is not supplied here.

One uncertain constraint and an explicit pulse design

Use the exact coupled reference above, $a_\lambda = S^{-1}Y_\lambda$, and

$$T_\lambda(w) = [C(\eta_\lambda(w, a_\lambda)) - C_*]/\lambda.$$

It satisfies $T_\lambda(0) = 0$, $T_0(w) = Lw$, and $J_\lambda = D_w T_\lambda(0) = L + O(\lambda)$. The moving reference matters: $T_\lambda = D_\lambda(w, a_\lambda) - D_\lambda(0, a_\lambda)$. Uniform inverse-function estimates give a fixed interval and actual curve

$$w_{\lambda,j}(s) = T_\lambda^{-1}(se_j), \qquad |s| \le S := \min\{s_0, \varepsilon/\lambda\}.$$

It has exactly one uncertain calibration, $C - C_* = \lambda s e_j$, all other constraints exact, and one full final record. Its tangent is $d_j + O(\lambda)$, $d_j = L^{-1}e_j$. If $(d_j)_x(d_j)_P \ne 0$, a smaller fixed neighbourhood preserves half both derivative magnitudes. Endpoints give

$$\epsilon_x\epsilon_P \ge \tfrac{1}{4}L_* P_* |(d_j)_x(d_j)_P| S^2.$$

The preceding upper bound still holds with three zero tolerances, giving matched order $\min\{1, (\varepsilon/\lambda)^2\}$. A zero leading entry calls for higher-order analysis, not an assertion of recovery. Every smooth projected curve patch has zero planar area despite positive coordinate risks.

All four column tests can hold in one fixed design. Choose pulse centres $\tau, 2\tau, 3\tau$ with small fixed $\tau > 0$. The output equation is $x^{(4)} + \alpha x'' + \beta x = 0$, where $\alpha = a/\mu + d/\nu$, $\beta = (ad - g^2)/(\mu\nu)$. In scaled initial-jet coordinates $(x, \tau\dot{x}, \tau^2\ddot{x}/2, \tau^3 x^{(3)}/6)$, $X(u) = x(\tau u)$ solves $X^{(4)} + \alpha\tau^2 X'' + \beta\tau^4 X = 0$. Integration on $[0, 3]$ makes the map $(x(\tau), x(2\tau), x(3\tau), \tau\dot{x}(\tau))$ an $O(\tau^2)$ perturbation of cubic Hermite interpolation. Its cardinal polynomials are

Output	Cardinal polynomial	Initial value	Initial derivative
1	$(u-2)(u-3)(3u-1)/4$	$-3/2$	$23/4$
2	$-(u-1)^2(u-3)$	3	-7
3	$(u-1)^2(u-2)/4$	$-1/2$	$5/4$
4	$(u-1)(u-2)(u-3)/2$	-3	$11/2$

A homogeneous cubic with roots $1, 2, 3$ and derivative zero at 1 vanishes, so this map is invertible. Substitution and differentiation verify the table. Initial momentum is μ/τ times the second entry. For pulse integrals $I_j > 0$, normalize calibration rows by $\sigma_j = -KI_j j\tau/M_j$ $(j = 1, 2, 3)$ and $\sigma_4 = -KcI_1/\tau$. Their narrow-pulse limits are those Hermite rows. With support half-width at most $h < \tau/4$, mean-value bounds on $(te(t))'/(j\tau)$ and $\tau e''(t)$ give a normalized matrix error $\|N - \mathcal{H}_\tau\| \le C_\tau h$. If $\|\mathcal{H}_\tau^{-1}\| C_\tau h < 1/2$, the inverse identity gives

$$\|N^{-1} - \mathcal{H}_\tau^{-1}\| \le 2\|\mathcal{H}_\tau^{-1}\|^2 C_\tau h.$$

Fix h so this is below half the smallest of the eight nonzero canonical entries. Both determinants and all column tests persist. Analytic observability supplies a fourth disjoint ordinary signal pulse. The widths are now positive and fixed; no shrinking-pulse physical limit enters the risk theorem. For another design the exact tests are the two cofactors $\mathrm{cof}_{j,1}(L), \mathrm{cof}_{j,2}(L)$ being nonzero.

Correlated uncertainty and the corrected quadratic response

Instead admit the fixed calibration line $C - C_* = \alpha r$, $|\alpha| \le \varepsilon$, $r = Le_y$, with this scalar normalization. It is not four independent error intervals. Its local reference-record curve is

$$w_\lambda(s) = T_\lambda^{-1}(sr), \quad |s| \le S, \qquad w_0(s) = se_y, \quad w_\lambda(0) = 0.$$

Bounded mixed derivatives give $\|w_\lambda(s) - se_y\| \le K_1 \lambda |s|$. For this compact common-record segment alone, optimal scalar error is half the coordinate range, so $R_x^{\mathrm{loc}} R_P^{\mathrm{loc}} \le L_* P_* K_1^2 \min(\lambda s_0, \varepsilon)^2$. This is a conditional local-fibre estimate, not a global estimator bound.

Set $N(\lambda, w) = C(\eta_\lambda(w, a_\lambda)) - C_*$ and $V(w) = \frac{1}{2}\partial_\lambda^2 N(0, w)$. Smooth inverse expansion gives

$$w_\lambda(s) = se_y - \lambda L^{-1} V(se_y) + O(\lambda^2 |s|).$$

The proposed nonzero tangent coefficient is actually zero. In the fixed linear cutoff region coupling/receiver reversal gives $G_{-\lambda}(-w, \eta) = G_\lambda(w, \eta)$ and $Y_{-\lambda} = Y_\lambda$. Uniqueness of the inverse chart implies $N(-\lambda, -w) = N(\lambda, w)$, hence V is even and $DV(0) = 0$. Moreover J_λ is even, so $J_\lambda = L + O(\lambda^2)$. The canonical tangent at the reference is $O(\lambda^2)$.

For completeness the actual finite-width quadratic coefficient is determined by terminal quadratures. In physical units put $x_w = e_x \Phi_t w$ and use expansion coefficients $u = u^{(0)} + \lambda u^{(1)} + \lambda^2 u^{(2)} + \cdots$. The first apparatus variations, with zero terminal data, are

$$u_j(t) = -\frac{K}{M_j} \int_t^T (r - t) f_j(r) x_w(r) dr, \quad k_j(t) = K \int_t^T f_j(r) x_w(r) dr,$$

$$h(t) = -\frac{Kc}{M_c} \int_t^T (r - t) f_1'(r) x_w(r) dr, \quad \ell(t) = Kc \int_t^T f_1'(r) x_w(r) dr.$$

The first receiver variation is independent of w, $z^{(1)}(t) = -Kc \int_0^t \Phi_{t-r} e_P f_1(r) dr$. It also occurs in the moving coupled reference. If $\xi = e_x z^{(1)}$, the reference's second terminal coefficients are the integrals $-K \int f_j \xi$, $-K \int (T - t) f_j \xi / M_j$ for probe momentum/position, and $-Kc \int f_1' \xi$, $-Kc \int (T - t) f_1' \xi / M_c$ for the clock. Thus $a_\lambda = \eta_c + \lambda^2 a_2 + O(\lambda^4)$, with a_2 the inverse free shear applied to that vector. This is its final preconditioned record; the reference's initial preparation stays exactly η_c.

Subtract the moving reference before second-order backward integration. The common $x^{(1)}$ terms cancel, leaving

$$\Delta \ddot{q}_j^{(2)} = -K f_j' h x_w / M_j, \qquad \Delta \dot{p}_s^{(2)} = -K(c f_1'' h x_w + \sum_j f_j' x_w u_j).$$

The resulting calibration coefficients are

$$V_j(w) = -\frac{K}{M_j}\int_0^T t f_j'(t)h(t)x_w(t)dt \quad (j=1,2,3),$$

$$V_4(w) = v_0 K\int_0^T (c f_1'' h x_w + \sum_j f_j' x_w u_j)dt + \frac{\ell(0)^2}{2M_c} + \sum_j \frac{k_j(0)^2}{2M_j}.$$

The squared momenta are necessary initial kinetic-energy terms. Length/energy units match the respective outputs; convert them to the fixed component units before applying L^{-1}. All first variations in these integrals are linear in w, so V is homogeneous quadratic, not just even. A free-reference calculation would retain incorrect constant forcing terms.

With $\beta = -\Pi_{x,P}L^{-1}V(e_y)$ this yields

$$\Pi_{x,P}w_\lambda(s) = \lambda s^2\beta + O(\lambda^2|s|), \qquad R_x^{\mathrm{loc}}R_P^{\mathrm{loc}} \le L_* P_* K_2^2(\lambda S^2 + \lambda^2 S)^2.$$

If both β entries are nonzero and each remainder is bounded by $M\lambda^2|s|$, then $S/\lambda \ge 2M/\min(|\beta_x|,|\beta_P|)$ permits comparison of $s=0$ with $s=S$, giving

$$R_x^{\mathrm{loc}}R_P^{\mathrm{loc}} \ge L_* P_* |\beta_x\beta_P|\lambda^2 S^4/16.$$

Opposite endpoints cancel the quadratic term and cannot prove this bound. Nonvanishing of both projected quadratures for the selected pulse family remains unproved. For S comparable to or smaller than coupling, higher coupling terms may compete; the displayed lower bound is not asserted.

Sources and consequence for STATE

The imported ingredients are smooth ODE dependence, contraction/inverse methods, Borsuk–Ulam, observability and Hermite interpolation. Their bounded source coverage and internal claim checks are retained in B48–B66, with the detailed terminal-response capsules in B65 and B66. The constrained charts, exact compensators, pulse matrices and quadratures above are the model derivations; no literature priority is claimed.

For STATE, positive preparation-relative risk, universal action selection, and identification with physical $\hbar$ remain different claims. Whole-shell, energy-constrained, local-curve and global-recovery hypotheses are explicit here. Further calibration stages are parked by action-scale obstructions; the conditional curvature test is retained without treating it as an active proof of necessity.